Piyush P. Jirwankar

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Education

University of California, Santa Cruz [2022 – Present]
Ph.D. candidate in the Department of Electrical and Computer Engineering
Advisor: Prof. Ricardo Sanfelice
GPA: 4.0/4.0

Indian Institute of Technology Bombay [2017 – 2022]
Interdisciplinary Dual Degree Program

- Bachelor of Technology in the Department of Aerospace Engineering
- Master of Technology in the Department of Systems and Control Engineering

GPA : 8.93/10

Research Interests

My research focuses on developing mathematically rigorous algorithms for robust estimation and safe control of nonlinear and hybrid systems evolving on manifolds. My work spans the analysis and control of hybrid systems on manifolds [2], [3], [4], [5], [8], [9], safe control design [2], [6], model predictive control [1], [7], and global attitude estimation [10], [12].

Publications and Preprints

[1] **Piyush P. Jirwankar**, Stefano Di Cairano, and Ricardo G. Sanfelice, *Input-to-State Stability of Model Predictive Control for Discretized Hybrid Dynamical Systems*, Under review at the American Control Conference (ACC), 2027.

[2] **Piyush P. Jirwankar**, Daniel E. Ochoa, Carlos A. Montenegro G., and Ricardo G. Sanfelice, *Multiple Control Barrier Functions for Geometric Hybrid Systems: Modeling and Safe Control Synthesis*, Under review at Springer MCSS, 2026.

[3] **Piyush P. Jirwankar**, Daniel E. Ochoa, and Ricardo G. Sanfelice, *Geometric Hybrid Dynamical Systems: Part I–Modeling and Stability*, Under review at IEEE Transactions on Automatic Control, 2026. (arxiv: 2609.17687).

[4] **Piyush P. Jirwankar**, Daniel E. Ochoa, and Ricardo G. Sanfelice, *Sufficient Conditions for Set Invariance Using Multiple Barrier Functions in Geometric Hybrid Dynamical Systems*, To appear in Proceedings of the 65th IEEE Conference on Decision and Control (CDC), 2026. (available online).

[5] Daniel E. Ochoa, **Piyush P. Jirwankar**, and Ricardo G. Sanfelice, *Solution Sets of Geometric Hybrid Systems: Topological Properties and Existence of Extremal Solutions*, To appear in Proceedings of the 65th IEEE Conference on Decision and Control (CDC), 2026. (available online).

[6] **Piyush P. Jirwankar**, Richard Montgomery, and Ricardo G. Sanfelice, *Hybrid Control for Robust and Global Obstacle Avoidance Via the Eikonal Equation*, To appear in Proceedings of the 65th IEEE Conference on Decision and Control (CDC), 2026. (available online).

[7] Himadri Basu, **Piyush P. Jirwankar**, Ricardo G. Sanfelice, Miguel Castroviejo-Fernandez, and Ilya Kolmanovsky, "A Hybrid Model Predictive Control Framework for Docking and Stabilization of Composite Rigid Spacecraft Dynamics," in *Proceedings of the AIAA SciTech Forum*, 2026. (doi:10.2514/6.2026-2855).

[8] **Piyush P. Jirwankar** and Ricardo G Sanfelice, "Geometric Hybrid Dynamical Systems on Intrinsic Manifolds: Part I–Framework and Hybrid Lyapunov Theorem," in *Proceedings of the 64th IEEE Conference on Decision and Control (CDC)*, IEEE, 2025, pp. 6593–6598. (doi:10.1109/CDC57313.2025.11313003).

[9] **Piyush P. Jirwankar**, Daniel E Ochoa, and Ricardo G Sanfelice, "Geometric Hybrid Dynamical Systems on Intrinsic Manifolds: Part II–Hybrid Invariance Principle," in *Proceedings of the 64th IEEE Conference on Decision and Control (CDC)*, IEEE, 2025, pp. 6599–6604. (doi:10.1109/CDC57313.2025.11312095).

[10] **Piyush P. Jirwankar**, Richard Montgomery, and Ricardo G. Sanfelice, "A Robust and Global Hybrid Complementary Filter on SO(3) using Morse Functions on RP3," in *Proceedings of the American Control Conference (ACC)*, 2025, pp. 4473–4478. (doi:10.23919/ACC63710.2025.11107772).

[11] Sebastian Henao-Garcia, Michael Kapteyn, Karen E Willcox, Marco Tezzele, Miguel Castroviejo-Fernandez, Taehyeun Kim, Michele Ambrosino, Ilya Kolmanovsky, Himadri Basu, **Piyush P. Jirwankar**, and Ricardo G. Sanfelice, "Digital-twin-enabled multi-spacecraft on-orbit operations," in *Proceedings of the AIAA SciTech Forum*, 2025, p. 1432. (doi:10.2514/6.2025-1432).

[12] **Piyush P. Jirwankar** and Ricardo G. Sanfelice, "Robust Global Hybrid Passive Complementary Filter on SO(2)," in *Proceedings of the 63rd IEEE Conference on Decision and Control (CDC)*, 2024, pp. 7946–7951. (doi:10.1109/CDC56724.2024.10886698).

[13] **Piyush P. Jirwankar**, Ravi N. Banavar, and Abhishek Abhishek, "Discrete-time optimal attitude control of swivelling biplane quadrotor," in *Proceedings of the AIAA SciTech Forum*, 2022, p. 1393. (doi:10.2514/6.2022-1393).

[14] Aniruddha R. Ranade et al., "Survey and Analysis of Payloads for Missions on PSLV's Orbital Platform," in *Proceedings of the AIAA SciTech Forum*, 2021, p. 0691. (doi:10.2514/6.2021-0691).

[15] Riya, Sanket Chirame, Anant A Joshi, Ram Milan Kumar Verma, Sumit Agrawal, Ravit Anand, Shrita Singh, Sanskriti Bhansali, **Piyush Jirwankar**, and Niket Parikh, "Closed-loop simulation for attitude control of nano-satellite," in *Advances in Small Satellite Technologies: Proceedings of 1st International Conference on Small Satellites*, Springer, 2020, pp. 87–97. (doi:10.1007/978-981-15-1724-2_9).

PhD Research

Modeling, Stability, and Safety of Geometric Hybrid Systems

- Modeled geometric hybrid systems on abstract manifolds, with and without inputs
- Derived a hybrid Lyapunov theorem and an invariance principle on manifolds by extending nonsmooth and set-valued analysis to geometric hybrid systems with state spaces defined on abstract manifolds
- Designed pointwise optimal safety filters and established conditions for measurable selections

Model Predictive Control of Hybrid Systems

- Established conditions for local input-to-state stability relative to a compact set for model predictive control (MPC) of hybrid dynamical systems represented through a discretized model
- Designed hysteretic hybrid MPC for robust spacecraft rendezvous and docking with impact dynamics

PDE-based Obstacle Avoidance for Ground Robots and Robotic Manipulators

- Designed obstacle avoidance using the Eikonal equation, a Hamilton–Jacobi PDE, ensuring safe, robust, global finite-time stabilization of a compact set for systems governed by integrator dynamics
- Extended Euclidean results to the Quanser QArm's configuration space for obstacle avoidance on manifolds

Robust and Global Attitude Estimation for Spacecraft and Quadrotors

- Developed a hybrid passive complementary filter to robustly overcome the topological obstruction that prevents global attitude estimation by continuous methods for spacecraft and quadrotors
- Achieved robustness to bounded disturbances and faster convergence than baseline EKF and MEKF

Research Internship | MERL

Contact Modeling for Warehouse Automation | Advisor: Scott Bortoff [Summer 2023]

- Built a finite-state machine for rigid-body contact and verified it using a bouncing-ball simulation in Modelica
- Validated the contact model theoretically through limit-cycle analysis of bouncing-ball dynamics

Technical Skills & Courses

Programming	Python, MATLAB, Simulink, Git/GitHub
Control	State Estimation, Stochastic & Networked Control, Process Control, Adaptive Control Theory, Distributed Optimization & Machine Learning, Advanced Feedback Control
Dynamics	Hybrid Dynamical Systems, Cyber Physical Systems, Geometric and Analytical Dynamics, Energy Methods in Structural Mechanics
Applied Math	Optimal Control, Convex Optimization, Applied Dynamical Systems
Aerospace	Flight Mechanics, Navigation & Guidance, Spacecraft Attitude Dynamics & Control

Additional Technical Projects

Model Predictive Control (MPC) Application [Spring 2021]
Course: Advanced Process Control, IIT Bombay

- Evaluated water-level control in a four-tank system under external disturbances, comparing the performance of PID, LQR, pole-placement, and linear MPC controllers

Attitude Estimation of a Quadrotor [Autumn 2020]
Course: State Estimation, IIT Bombay

- Estimated quadrotor attitude from simulated IMU measurements using Kalman and extended Kalman filters, and investigated the explicit complementary filter
- Assessed filter performance through innovations, estimation errors, covariance-matrix spectral radii, state RMSE, and normalized estimation error squared (NEES)

Adaptive Control Application [Autumn 2020]
Course: Adaptive Control Theory, IIT Bombay

- Applied adaptive **integrator backstepping** to trajectory tracking for a two-link manipulator
- Implemented adaptive feedback for spacecraft **attitude stabilization** under unknown inertia

Mentorship

2026 Science Internship Program | UC Santa Cruz [Summer 2026]
Research mentorship for high school interns interested in research projects at UC Santa Cruz

- Mentored four high school students implementing and validating Eikonal-based obstacle avoidance on the Quanser QArm, designing tasks to support collaboration and algorithm implementation
- Taught MATLAB/Simulink, calculus, ordinary and partial differential equations (ODEs/PDEs), numerical methods for solving ODEs/PDEs, state-space modeling, and forward kinematics

Student Satellite Program | IIT Bombay [2018 – 2022]
Student-led initiative to advance satellite and space technology at IIT Bombay

- Led the GNC subsystem and employed systems engineering principles to chalk out strategies for GNC for autonomous rendezvous and docking of two 6U spacecraft
- Managed annual recruitment and mentored new members on guidance, navigation, and control projects

Undergraduate Mentorship | IIT Bombay [2019 – 2022]

- Served as Institute Student Mentor to eight first-year undergraduates, supporting their transition to college life and guiding their academic and extracurricular pursuits at IIT Bombay
- Served as Department Academic Mentor to six sophomores and two seniors facing academic challenges, providing guidance to help them improve their academic performance

Teaching and Service

Teaching

– Systems & Control Lab | IIT Bombay [Jan 2022 – Apr 2022]
  - Conducted lab sessions on Markov Chain Monte Carlo (MCMC) algorithms, helping students solve simulation problems using Gibbs sampling and the Metropolis–Hastings algorithm
  - Helped students apply simulated annealing to global optimization problems

– Lectures on Differential Geometry | IIT Bombay [Aug 2021 – Sep 2021]
  - Delivered differential geometry lectures to a group of enthusiastic undergraduate students
  - Prepared teaching materials and conducted differential geometry tutorials covering topological manifolds, tangent and cotangent bundles, as well as Lie groups and their actions

Reviewer
Automatica, IEEE Conference on Decision and Control, American Control Conference